A
PROJECT REPORT
ON

(Skill Assessment and Career Guidance Website)

SUBMITTED BY

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And

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ACADEMIC YEAR 2026-27

T.Y. B.C.A. SEM - 5

UNDER THE GUIDANCE OF

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CERTIFICATE OF PROJECT COMPLETION

This is to certify that the project entitled
"Skill assessment and career recommendation system"
is a bonafide work carried out by
Prathana Dama N. (24CS002UG04214)
and **Prachi Kanani P. (24CS002UG04335)**
as a partial fulfillment of the requirements
for the B.C.A. Semester 5 during the
Academic Year 2026-2027.

The project report meets the academic requirements for the project work prescribed by Saurashtra University, Rajkot.

EXAMINER NAME & SIGN	**PRINCIPAL SIGN**
Mr. Yash Kotecha	/ /2026
PROJECT GUIDE	**EXAM DATE**

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ABSTRACT

NextPath is a Career Guidance and Skill Assessment web application which aims to help students identify suitable career opportunities based on their interests, aptitude and skills. Presently, students often face difficulty in choosing the right career due to lack of proper guidance and awareness about various career options.

In the proposed system, students can take a skill assessment test, and based on their responses, the system analyzes their strengths and recommends suitable career paths. The application also provides detailed information about different careers and related courses, enabling students to make informed decisions about their future.

ACKNOWLEDGEMENT

The feeling of gratefulness to anyone's help directly arises from the bottom of the heart. A small but important and timely help can prove to be a milestone in one's life.

This project is dedicated to all the people whom we met, took guidance from, talked to, interviewed and learned something from. On this occasion we sincerely thank all of them while submitting this project report.

We would like to thank **Mr. Yash Kotecha** for giving time-to-time valuable guidance during the project, and for guiding us in taking our curriculum decisions and in choosing, initiating and carrying out the project.

We have always felt the invisible help of the Almighty; without the blessings of the Almighty, we could not have succeeded.

Prathana Dama

Prachi Kanani

PROJECT PROFILE

DEVELOPERS	
Name	Enrollment Number
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PROJECT DETAILS	
Name of Project	NextPath
Project Title	Skill Assessment and Career Recommendation System
Platform	Front End: HTML5, CSS3, JavaScript Back End: PHP, MySQL
Duration	3 Months
Team Size	2
GUIDE INFORMATION	
Name of Guide	Mr. Yash Kotecha

1. SDLC OVERVIEW



Figure 1.1: Software Development Life Cycle of NextPath

2. REQUIREMENT GATHERING AND ANALYSIS

2.1 Organization Details

1. Name of the Organization

NextPath Career Solutions Pvt. Ltd.

2. Brief Detail of the Organization

NextPath Career Solutions Pvt. Ltd. is an educational technology (EdTech) organization dedicated to helping students and career seekers make informed career decisions. The organization focuses on providing career guidance through aptitude assessment, interest evaluation and career exploration. Its mission is to bridge the gap between students' abilities and suitable career opportunities by offering a digital platform that delivers accurate career recommendations and detailed career information.

2.2 Meetings

I. Meeting with Project Coordinator

1. Name of Project Coordinator: Mr. Niraj Mehta

2. Project Coordinator's requirements in his words:

- Develop a secure web-based **Career Guidance and Skill Assessment System**.
- Allow users to **register and log in securely**.
- Conduct online **skill assessments** and analyze user responses.
- Generate **career recommendations** based on assessment performance.
- Display **career and course information** for users.
- Provide an **admin panel** to manage users, questions, categories, careers and courses.
- Ensure the system is **user-friendly, responsive and easy to maintain**.

II. Meeting with Users

1. Users:

- College Students
- Fresh Graduates
- Career Seekers

2. User requirements in their words:

- Simple and secure **registration and login** process.

- Easy-to-use **skill assessment** with clear questions.
- Career recommendations based on **assessment performance and skills**.
- Detailed information about **careers and related courses**.
- Easy access to **assessment results and history**.
- A **user-friendly and responsive** interface accessible on different devices.
- Secure storage of **personal information and assessment records**.

2.3 Data which will be Input into the System

- Student Registration Details
- Login Credentials
- Personal Information
- Educational Information
- Profile Update Information
- Interest Assessment Responses
- Career Assessment Responses
- Aptitude Test Responses
- Skill Assessment Responses
- Interests, Skills and Hobbies
- Career and Course Data entered by Administrator
- Assessment Questions and Categories managed by Administrator

2.4 Data which will be Output from the System

- Student Profile
- Assessment Score and Performance
- Career Analysis Report
- Career Assessment Results
- Career Information and Career Details
- Course Information related to Career Options
- Career Categories and Available Career Options

2.5 Type of Project

- Web-Based Online Application

2.6 Method of Collecting Data

- Interviews
- Questionnaires
- Career Research
- Internet Research
- Books and Educational Resources

3. SYSTEM REQUIREMENT SPECIFICATIONS

3.1 Introduction

3.1.1 Purpose

NextPath – Career Guidance and Skill Assessment System is a web-based application designed to assist students in identifying suitable career options based on their interests, aptitude and skills. The system provides an online assessment, analyzes the user's responses and recommends appropriate career paths along with detailed career information to help students make informed career decisions.

3.1.2 Intended Audience and Reading Suggestions

The intended audience of this document includes:

- Students
- Career Seekers
- Project Guide
- Developers
- System Administrator

3.1.3 Scope

The proposed system is a web-based **Career Guidance and Skill Assessment platform** that enables students and career seekers to register, complete an online skill assessment, view their assessment results and receive career recommendations based on their performance. The system also allows users to explore detailed information about various careers and related courses. An administrative panel is provided for managing users, assessment questions, categories, career information and course data.

3.2 Overall Description

3.2.1 Product Function

(i) Admin Module:

- Secure Admin Login
- Manage Student Records
- Manage Assessment Questions and Categories
- Add / Edit / Delete Career and Course Information
- View Assessment Records and Statistics

- Manage System Data

(ii) User Module:

- User Registration
- Secure Login
- Update Profile
- Take Skill Assessment
- View Assessment Results and Skill Profile
- View Recommended Careers
- Explore Career and Course Details
- View Assessment History
- Logout

3.2.2 User Classes and Characteristics

(i) User Classes:

- Administrator
- User / Student

(ii) User Characteristics:

(a) Administrator

- Manages assessment questions.
- Manages career and course information.
- Monitors student records and assessment data.
- Views system statistics and maintains system data.

(b) User / Student

- Can register and log in securely.
- Can complete the skill assessment.
- Can view assessment results and recommended careers.
- Can explore career and course information.
- Can update personal profile and view assessment history.
- Basic knowledge of using a web browser is sufficient.

3.2.3 Operating Environment

The application operates through an internet connection.

Hardware:

- RAM

Operating System:

- Windows

3.2.4 Design and Implementation Constraints

- English language interface.
- Internet connection required.
- Users must have basic computer or smartphone knowledge.
- Only administrators can modify career and assessment data.
- Secure login is required for all users.

3.2.5 Assumptions and Dependencies

- Administrator account already exists.
- Users have access to a **web browser and required connection**.
- PHP and MySQL are properly configured.
- Assessment questions, categories, career and course information are maintained by the administrator.
- Users are expected to provide **accurate information and responses** during registration and assessment.

3.3 External Interface Requirement

3.3.1 User Interface

The application provides a simple and interactive interface. Users can:

- Register
- Login
- Update Profile
- Take Assessment Test
- Submit Answers
- View Recommended Careers
- Explore Career and Course Details
- View Assessment History
- Logout

If incorrect login credentials are entered, an appropriate error message is displayed.

3.3.2 Software Interface

Component	Technology
Operating System	Windows
Front End	HTML5, CSS3, Bootstrap, JavaScript
Back End	PHP
Database	MySQL
Development Tool	Visual Studio Code (Version 1.69.0)
Web Server	Apache (XAMPP)

3.4 System Features

- User Registration
- Secure Login
- Profile Management
- Skill Assessment
- Assessment Result and Skill Profile
- Career Recommendation
- Career and Course Information
- Assessment History
- Admin Dashboard
- Question and Category Management
- Career and Course Management
- Student and Assessment Management
- System Statistics

3.5 Other Non-functional Requirements

3.5.1 Performance Requirement

- Fast response time.
- Smooth navigation between pages.
- Simple and user-friendly interface.
- Efficient database operations.

3.5.2 Security Requirement

- Secure user authentication and session management.

- Password protection and secure storage.
- Role-based access for Administrator and Student.
- Protection of user and assessment data.
- Input validation and secure database operations.

3.5.3 Safety Requirement

- Regular database backup.
- Protection against unauthorized data modification.
- Proper validation and handling of user data.
- Secure storage of important information.
- Error handling to prevent system failure.

3.6 Feasibility Study

A feasibility study evaluates whether the proposed system is practical, beneficial and capable of meeting user requirements within available resources.

3.6.1 Operational Feasibility

The **NextPath** system is simple and user-friendly. It allows students to complete career assessments, view results and explore career and course information. The administrative module helps manage students, assessments, careers and courses efficiently.

3.6.2 Technical Feasibility

The **NextPath** system uses available technologies such as **HTML, CSS, JavaScript, PHP and MySQL**. These technologies are suitable for developing and maintaining the web application.

3.6.3 Economic Feasibility

The project has **low development cost** as it uses open-source technologies and freely available tools. It also reduces manual effort in career assessment and information management.

3.6.4 Schedule Feasibility

The **NextPath** project can be completed within the allotted academic schedule. The selected technologies support timely development, testing and implementation.

4. SYSTEM ANALYSIS AND MODELING

4.1 Use Case Diagram

(a) Administrator Side

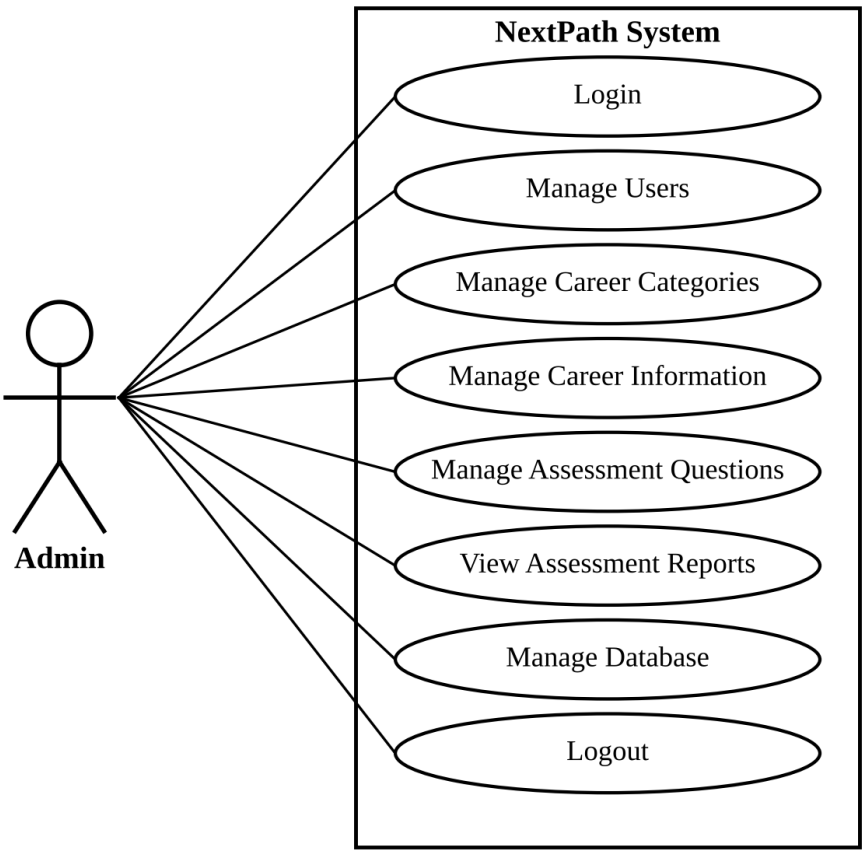


Figure 4.1: Use Case Diagram – Administrator

(b) User / Student Side

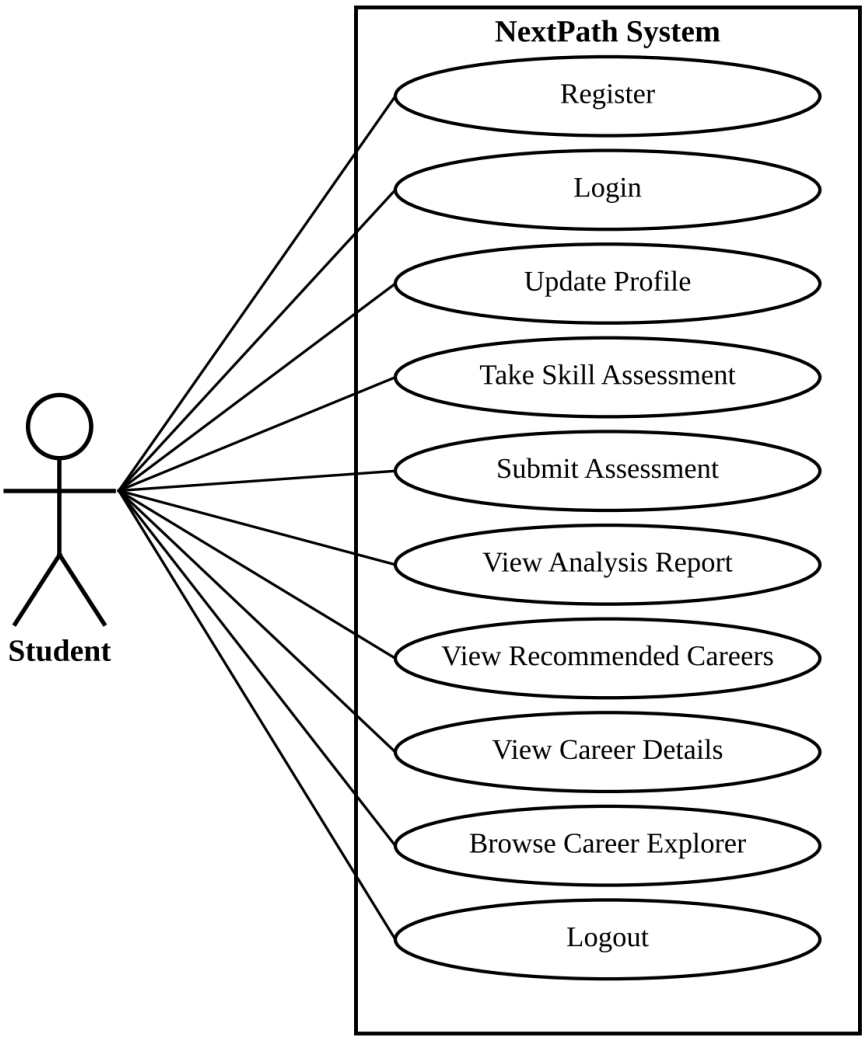


Figure 4.2: Use Case Diagram – Student

4.2 Normalization

Normalization is the process of organizing the columns (attributes) and tables (relations) of a relational database to reduce data redundancy and improve data integrity.

(A) De-normalized Form

User ID	User Name	Email	Education	College	Interests	Skills	Category
1	poppy	poppy@gmail.com	Undergraduate	CZMG	Photography	Editing	Science
2	moppy	moppy@gmail.com	Undergraduate	CZMG	Photography	Editing	Science

Career	Course	Skill	Question	Selected Answer	Assessment Score
Research Scientist	B.Sc in Physics / Chemistry / Biology	Analytical Thinking	Which activity interests you most?	A	25
Lab Technician	M.Sc in a Core Science Discipline	Analytical Thinking	What excites you most about a school project?	B	25
Software Engineer	B.E. / B.Tech	Problem Solving	Which activity interests you most?	A	25

(B) First Normal Form (1NF)

1) Table Name: tbl_user

Primary Key: id (Auto Increment)

id	full_name	email	mobile	gender	education_level
1	NEXTPath Admin	admin@nextpath.com	98XXXXXXXX	Other	Graduate
3	poppy	poppy@gmail.com	91XXXXXXXX	Female	Undergraduate

college_name	interests	skills	hobbies	role
NEXTPath HQ	NULL	NULL	NULL	admin
CZMG	Photography	Editing	Sports	student

2) Table Name: tbl_categories

Primary Key: id (Auto Increment)

id	name	description
1	Science	Explore careers built on scientific inquiry, experimentation and discovery.
2	Accounting & Finance	Careers involving financial records, analysis, auditing and planning.

id	name	description
3	Banking	Careers in retail and corporate banking, credit and financial services.
4	Nursing & Healthcare	Careers focused on patient care, treatment and health services.

3) Table Name: tbl_careers

Primary Key: id (Auto Increment)

id	category_id	career_name	relevance
1	1	Research Scientist	1
2	1	Lab Technician	0.9
3	1	Science Journalist	0.75
4	2	Accountant	1
5	2	Financial Analyst	0.95
6	2	Tax Consultant	0.85

4) Table Name: tbl_courses

Primary Key: id (Auto Increment)

id	category_id	course_name	description
1	1	B.Sc in Physics / Chemistry / Biology	A foundational science degree that opens the door to research, teaching and specialised M.Sc study.
2	1	M.Sc in a Core Science Discipline	Advanced specialisation for students aiming at research or a doctoral pathway.
3	2	B.Com (Accounting & Finance)	Covers financial accounting, auditing and taxation fundamentals.
4	2	Chartered Accountancy (CA) / CMA	Professional certification for accounting, audit and cost management careers.

5) Table Name: tbl_category_skills

Primary Key: id (Auto Increment)

id	category_id	skill_name	weight
1	1	Analytical Thinking	1
2	2	Analytical Thinking	1
3	3	Analytical Thinking	1
4	4	Communication	1
5	5	Creativity	1
6	6	Leadership	1

6) Table Name: tbl_questions

Primary Key: id (Auto Increment)

id	category_id	question_text	option_a	option_b	option_c	option_d	correct_answer
1	1	Which activity interests you most?	Running a controlled experiment	Preparing financial reports	Designing a marketing campaign	Managing a hotel front desk	A
2	1	What excites you most about a school project?	Testing a hypothesis with real data	Sketching a poster for the event	Interviewing people for a story	Organising a sports tournament	A
3	2	Which activity would you prefer?	Preparing financial reports	Designing websites	Treating patients	Creating animations	A

7) Table Name: tbl_assessments

Primary Key: id (Auto Increment)

id	user_id	total_questions	correct_answers	overall_score	completed_at
2	3	40	10	25	2026-09-05 11:42

8) Table Name: tbl_assessment_answers

Primary Key: id (Auto Increment)

id	assessment_id	question_id	selected_answer	is_correct	score
41	2	40	A	1	1
42	2	27	B	0	0
43	2	3	B	0	0
44	2	36	B	0	0
45	2	23	A	1	1

9) Table Name: tbl_settings

Primary Key: setting_key

setting_key	setting_value
assessment_length	40

(C) Second Normal Form (2NF)

In 2NF every non-key attribute is fully dependent on the primary key. Repeating career, course, skill, question and answer data is moved into separate tables that are linked through foreign keys.

1) Table Name: tbl_user

Primary Key: id (Auto Increment)

id (PK)	full_name	email	mobile	gender	education_level
1	NEXTPath Admin	admin@nextpath.com	98XXXXXXXX	Other	Graduate
3	poppy	poppy@gmail.com	91XXXXXXXX	Female	Undergraduate

college_name	interests	skills	hobbies	role
NEXTPath HQ	NULL	NULL	NULL	admin
CZMG	Photography	Editing	Sports	student

2) Table Name: tbl_categories

Primary Key: id (Auto Increment)

id (PK)	name	description	created_at	updated_at
1	Science	Explore careers built on scientific inquiry, experimentation and discovery.	2026-07-10	2026-07-10
2	Accounting & Finance	Careers involving financial records, analysis, auditing and planning.	2026-07-10	2026-07-10
3	Banking	Careers in retail and corporate banking, credit and financial services.	2026-07-10	2026-07-10
4	Nursing & Healthcare	Careers focused on patient care, treatment and health services.	2026-07-10	2026-07-10

3) Table Name: tbl_careers

Primary Key: id (Auto Increment) **Foreign Key:** category_id

id (PK)	category_id (FK)	career_name	description	relevance
1	1	Research Scientist	Designs and runs experiments to advance scientific knowledge.	1
2	1	Lab Technician	Supports experiments and maintains laboratory equipment.	0.9
3	1	Science Journalist	Communicates scientific findings to the public.	0.75
4	2	Accountant	Prepares and audits financial records for organisations.	1
5	2	Financial Analyst	Analyses financial data to guide investment decisions.	0.95

id (PK)	category_id (FK)	career_name	description	relevance
6	2	Tax Consultant	Advises individuals and businesses on tax planning.	0.85

4) Table Name: tbl_courses

Primary Key: id (Auto Increment) **Foreign Key:** category_id

id (PK)	category_id (FK)	course_name	description
1	1	B.Sc in Physics / Chemistry / Biology	A foundational science degree that opens the door to research, teaching and specialised M.Sc study.
2	1	M.Sc in a Core Science Discipline	Advanced specialisation for students aiming at research or a doctoral pathway.
3	2	B.Com (Accounting & Finance)	Covers financial accounting, auditing and taxation fundamentals.
4	2	Chartered Accountancy (CA) / CMA	Professional certification for accounting, audit and cost management careers.
5	3	B.Com with Banking & Insurance	Focused commerce degree covering banking operations and financial services.
6	3	Diploma in Banking & Finance	A shorter, practice-oriented path into retail and corporate banking roles.

5) Table Name: tbl_category_skills

Primary Key: id (Auto Increment) **Foreign Key:** category_id

id (PK)	category_id (FK)	skill_name	weight
1	1	Analytical Thinking	1
2	2	Analytical Thinking	1
3	3	Analytical Thinking	1
4	4	Communication	1

6) Table Name: tbl_questions

Primary Key: id (Auto Increment) **Foreign Key:** category_id

id (PK)	category_id (FK)	question_text	option_a	option_b	option_c	option_d	correct_answer
1	1	Which activity interests you most?	Running a controlled experiment	Preparing financial reports	Designing a marketing campaign	Managing a hotel front desk	A

id (PK)	category_id (FK)	question_text	option_a	option_b	option_c	option_d	correct_answer
2	1	What excites you most about a school project?	Testing a hypothesis with real data	Sketching a poster for the event	Interviewing people for a story	Organising a sports tournament	A
3	2	Which activity would you prefer?	Preparing financial reports	Designing websites	Treating patients	Creating animations	A
4	2	What sounds most satisfying to you?	Reconciling a company's accounts	Coaching a football team	Writing a news article	Designing a building facade	A

7) Table Name: tbl_assessments

Primary Key: id (Auto Increment) **Foreign Key:** user_id

id (PK)	user_id (FK)	total_questions	correct_answers	overall_score	completed_at
2	3	40	10	25	2026-09-05 11:42

8) Table Name: tbl_assessment_answers

Primary Key: id (Auto Increment) **Foreign Key:** assessment_id, question_id

id (PK)	assessment_id (FK)	question_id (FK)	selected_answer	is_correct	score
41	2	40	A	1	1
42	2	27	B	0	0
43	2	3	B	0	0
44	2	36	B	0	0
45	2	23	A	1	1
46	2	29	A	1	1
47	2	38	C	0	0
48	2	15	D	0	0

9) Table Name: tbl_settings

Primary Key: setting_key

setting_key (PK)	setting_value
assessment_length	40

4.3 E-R Diagram

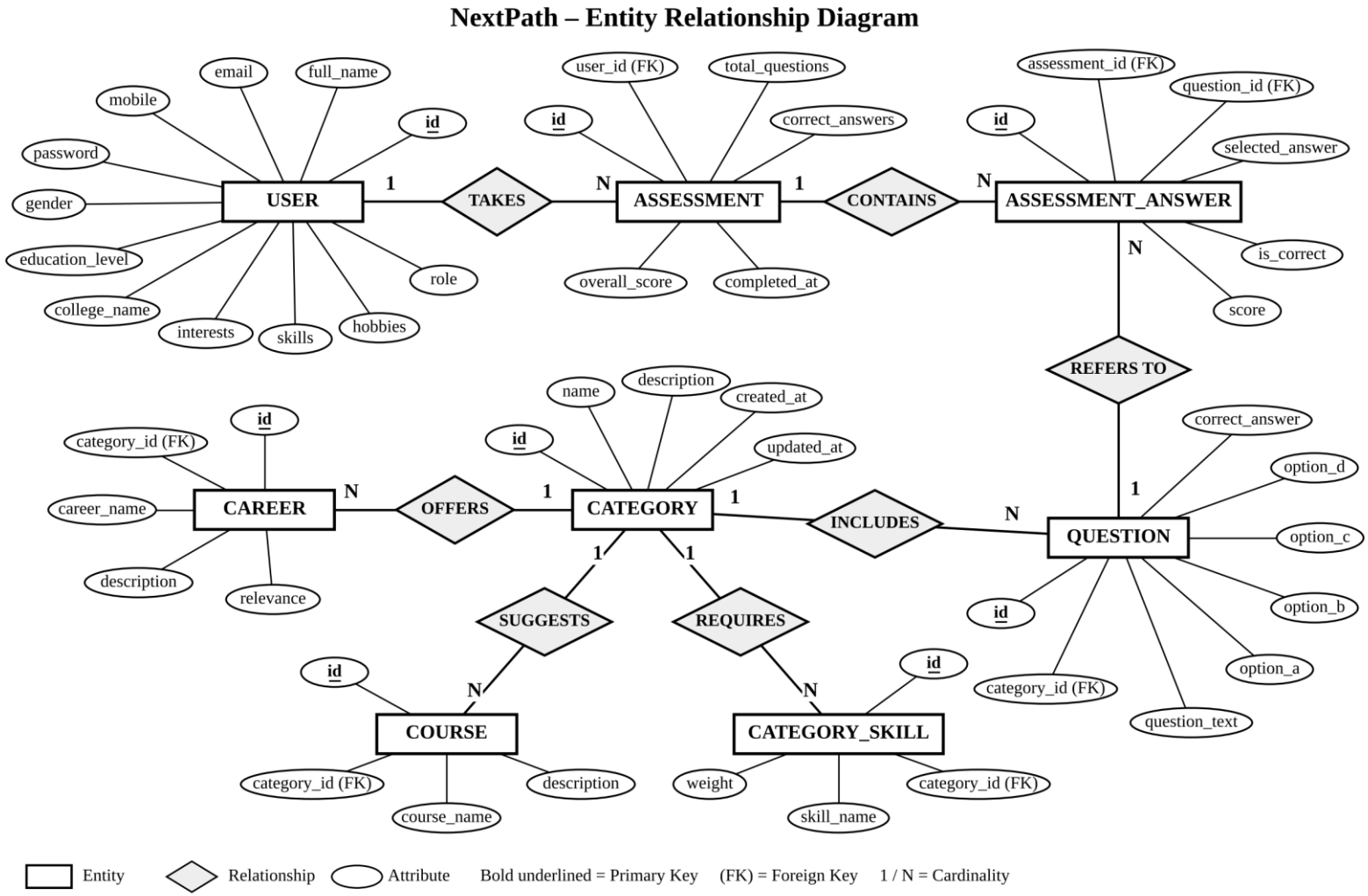


Figure 4.3: E-R Diagram of NextPath (Chen Notation)

4.4 Data Dictionary

1. tbl_user

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the user
full_name	VARCHAR		Full name of the user
email	VARCHAR		Email address used for login; must be unique
mobile	VARCHAR		Mobile number (up to 10 digits)
password	VARCHAR		One-way hashed password
gender	VARCHAR		Male / Female / Other (optional)
education_level	VARCHAR		High School, Higher Secondary, Undergraduate, Graduate or Postgraduate (optional)
college_name	VARCHAR		School / college name (optional)
interests	VARCHAR		Interests of the student (optional)
skills	VARCHAR		Skills of the student (optional)
hobbies	VARCHAR		Hobbies of the student (optional)
role	ENUM		'student' or 'admin'

2. tbl_categories

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the category
name	VARCHAR		Career category name (28 categories)
description	TEXT		Short description shown on the home page and explorer
created_at	DATETIME		Date and time the record was created
updated_at	DATETIME		Date and time the record was last updated

3. tbl_questions

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the question
category_id	INT	FK	Category to which the question belongs (tbl_categories.id)
question_text	VARCHAR		Question statement

Field Name	Data Type	Key	Description
option_a – option_d	VARCHAR		Four multiple-choice options
correct_answer	VARCHAR		Correct option used for scoring

4. tbl_assessments

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the assessment attempt
user_id	INT	FK	Student who took the assessment (tbl_user.id)
total_questions	INT		Number of questions in the attempt
correct_answers	INT		Number of correct answers
overall_score	DECIMAL		Correct answers ÷ total questions × 100
completed_at	DATETIME		Time of the attempt, shown in history and log

5. tbl_assessment_answers

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the stored answer
assessment_id	INT	FK	Assessment attempt to which the answer belongs (tbl_assessments.id)
question_id	INT	FK	Question that was answered (tbl_questions.id)
selected_answer	VARCHAR		Option chosen by the student
is_correct	TINYINT		1 if the selected answer is correct, otherwise 0
score	INT		Score awarded for the answer

6. tbl_careers

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the career
category_id	INT	FK	Category of the career (tbl_categories.id)
career_name	VARCHAR		Name of the career (84 careers)
description	TEXT		Short description of the career
relevance	DECIMAL		Weight from 0 to 1 used in the career match

7. tbl_courses

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the course
category_id	INT	FK	Category for which the course is suggested (tbl_categories.id)
course_name	VARCHAR		Suggested course or degree (56 courses)
description	TEXT		Short description of the course

8. tbl_category_skills

Field Name	Data Type	Key	Description
id	INT	PK	Unique identifier of the mapping
category_id	INT	FK	Category contributing to a skill (tbl_categories.id)
skill_name	VARCHAR		Skill of the Skill Profile (e.g. Communication)
weight	DECIMAL		Weight of the category in the skill calculation

4.5 Functional and Behavior Modeling

4.5.1 Context Diagram (0-Level Diagram)

The context diagram shows the complete NextPath system as a single process (0.0) and the data that flows between the system and its two external entities – the Student and the Administrator.

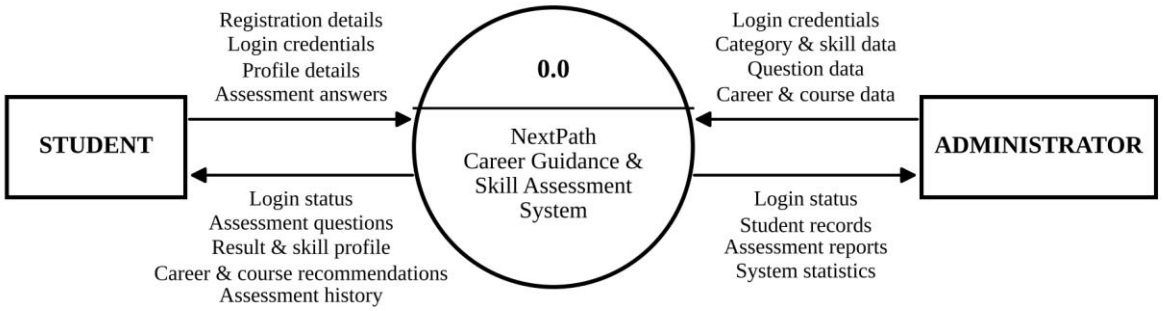


Figure 4.4: Context Diagram (0-Level DFD)

Symbols used in the DFD:

Symbol	Name	Meaning
Rectangle	External Entity	A person outside the system that sends or receives data (Student, Administrator)
Circle	Process	Work done on data, numbered 0.0, 1.0, 2.0 …
Open-ended rectangle	Data Store	Place where data is stored; shown as D1, D2 … with the table name
Arrow	Data Flow	Direction in which the named data moves

4.5.2 First Level Diagram (1-Level Diagram)

In the 1-level DFD the single process of the context diagram is broken into its main sub-processes. Each **data store is labelled D1 to D8**; every data store is one table of the NextPath database, as listed below. The same numbers are used in both the Administrator and Student diagrams.

Data Store	Table Name	Data Stored
D1	tbl_user	Student and administrator accounts and profiles
D2	tbl_categories	Career categories
D3	tbl_questions	Assessment questions with options and correct answer
D4	tbl_careers	Careers under each category
D5	tbl_courses	Courses suggested for each category
D6	tbl_category_skills	Skill weights of each category
D7	tbl_assessments	Assessment attempts and scores
D8	tbl_assessment_answers	Answers given in each attempt

(a) Administrator

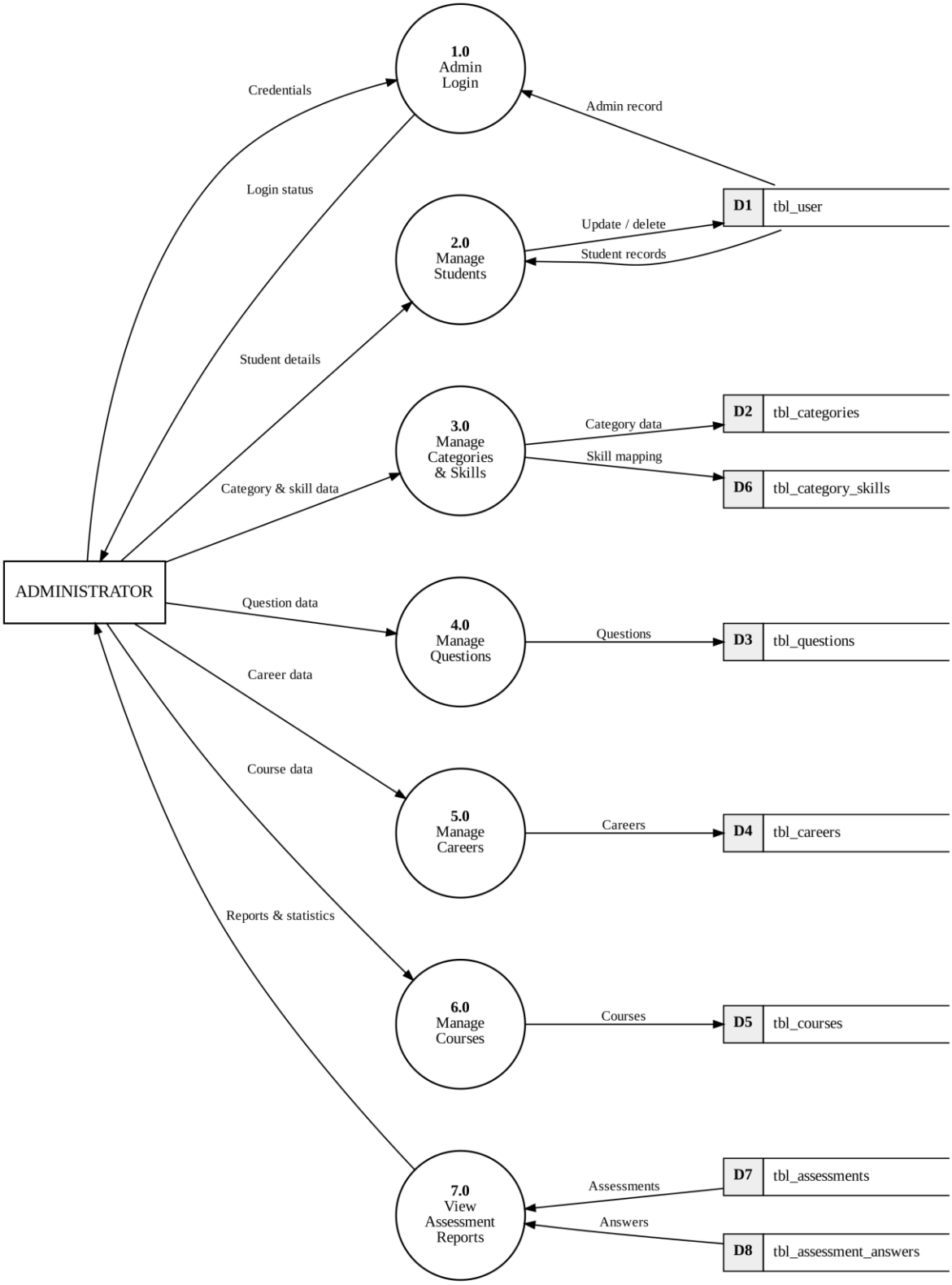


Figure 4.5: 1-Level DFD – Administrator

(b) User / Student

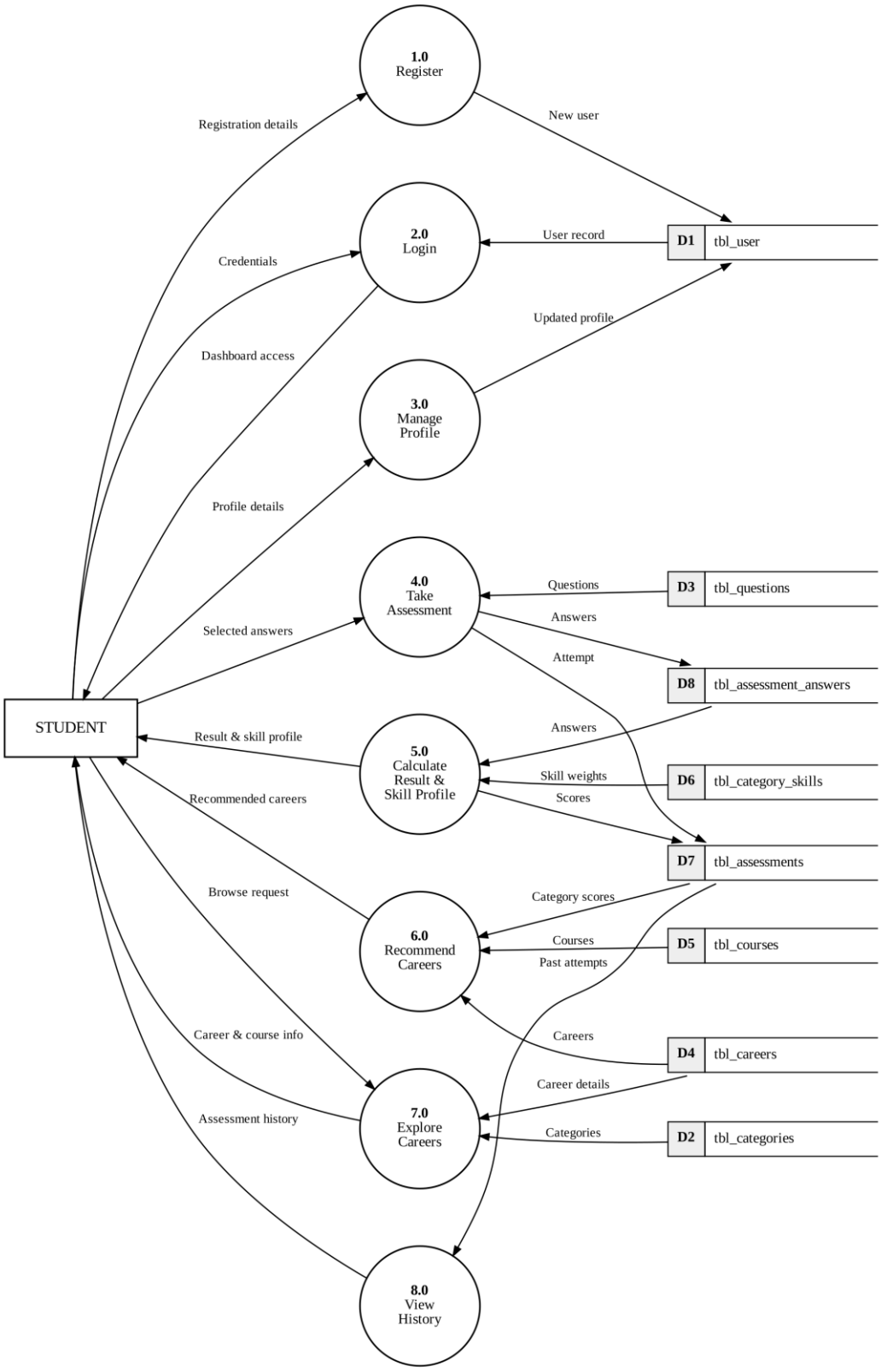


Figure 4.6: 1-Level DFD – Student

4.6 Gantt Chart

The Gantt chart below compares the expected time (blue) and the actual time (green) taken for each phase of the project from July to October 2026.

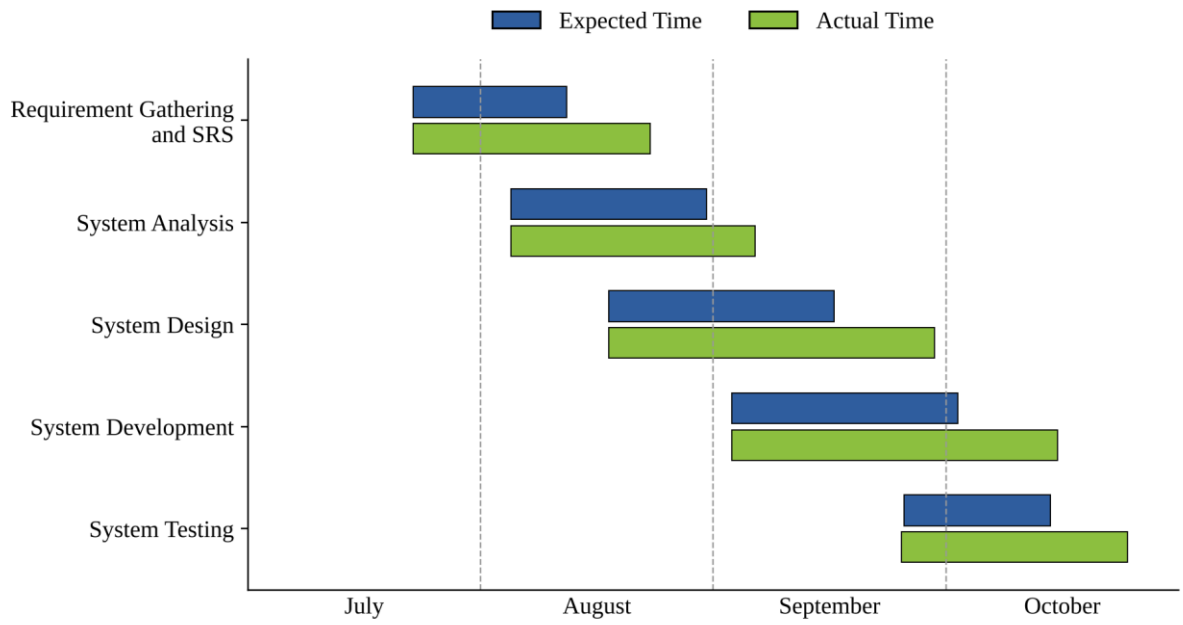


Figure 4.7: Gantt Chart

5. TEST CASES

Invalid Login					
TEST CASE No.: 1			**TEST CASE DESIGN DATE:** 20/07/2026		
TEST TITLE: Invalid Login Test			**TEST CASE EXECUTION DATE:** 05/08/2026		
DESCRIPTION: In this test case the invalid login handling of the Login module is tested.					
PRECONDITION: Login page must be working.					
Step No.	**Test Step**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
1	Submit empty email and password	Blank	Messages "Please enter a valid email address." and "Please enter your password." are displayed.	Same as expected	PASS
2	Enter a wrong password	Valid email, wrong password	Message "Incorrect email or password." is displayed.	Same as expected	PASS
3	Enter an unregistered email	Unknown email	Message "Incorrect email or password." is displayed and no session is created.	Same as expected	PASS

Login					
TEST CASE No.: 2			**TEST CASE DESIGN DATE:** 15/08/2026		
TEST TITLE: Login Test			**TEST CASE EXECUTION DATE:** 25/08/2026		
DESCRIPTION: In this test case the Login module is tested.					
PRECONDITION: Login page must be working.					
Step No.	**Test Step**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
1	Open login.php	–	Login form with Email and Password fields and Show toggle is displayed.	Same as expected	PASS
2	Login with valid student credentials	Student email and password	Session is created and student dashboard opens.	Same as expected	PASS
3	Login with valid admin credentials	Admin email and password	Session is created and admin dashboard opens.	Same as expected	PASS
4	Open login.php while logged in	–	User is redirected to the dashboard according to their role.	Same as expected	PASS

Skill Assessment Attempt					
TEST CASE No.: 3			**TEST CASE DESIGN DATE:** 31/08/2026		
TEST TITLE: Skill Assessment Attempt Test			**TEST CASE EXECUTION DATE:** 05/09/2026		
DESCRIPTION: Verify the skill assessment attempt with mixed-category questions.					
PRECONDITION: Student must be logged in and the question bank must contain questions.					
Step No.	**Test Step**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
1	Click Start Assessment	–	First question is displayed in the one-question interface.	Same as expected	PASS
2	Select an option and click Next	Any option	Next question is displayed and the progress bar advances.	Same as expected	PASS
3	Click Previous	–	Previous question is displayed with the earlier selection retained.	Same as expected	PASS
4	Observe the question order	–	Questions from different categories are mixed rather than grouped.	Same as expected	PASS

Assessment Submission and Score Calculation					
TEST CASE No.: 4			**TEST CASE DESIGN DATE:** 15/09/2026		
TEST TITLE: Assessment Submission and Score Calculation Test			**TEST CASE EXECUTION DATE:** 30/09/2026		
DESCRIPTION: Verify assessment submission and calculated scores.					
PRECONDITION: All assessment questions must be attempted.					
Step No.	**Test Step**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
1	Submit the assessment	All answers selected	Each answer is stored with its correctness in tbl_assessment_answers.	Same as expected	PASS
2	Check overall score	30 correct out of 40	Overall score is calculated as 75%.	Same as expected	PASS
3	Check category score	3 correct out of 4	Category score is calculated as 75%.	Same as expected	PASS
4	Check skill profile and strengths	Category scores	Skill scores are calculated and strongest areas are ranked.	Same as expected	PASS

6. SCREENSHOTS

6.1 Home Page

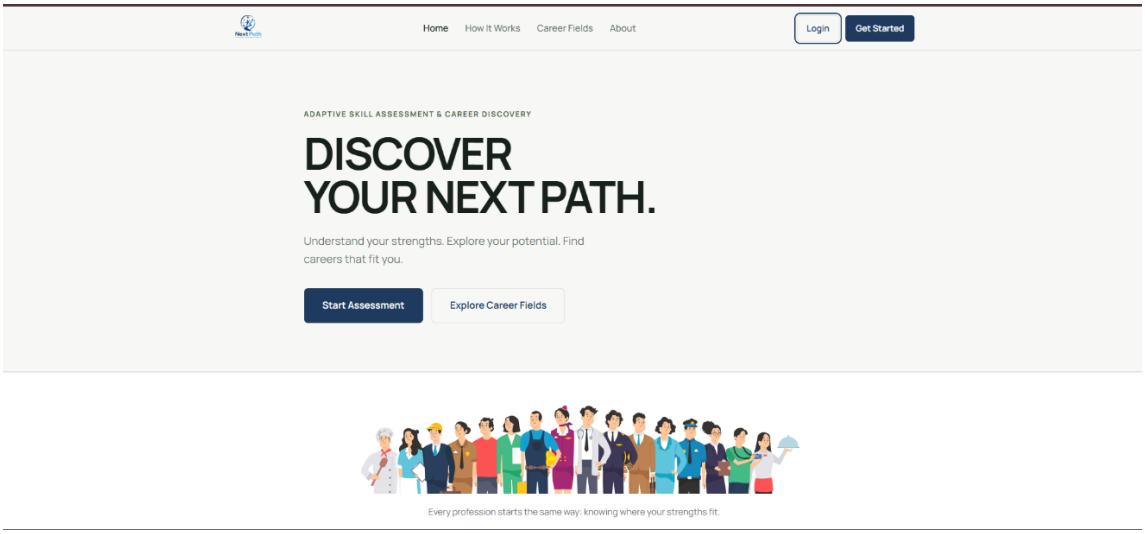


Figure 6.1: Home Page

6.2 Student Registration Page

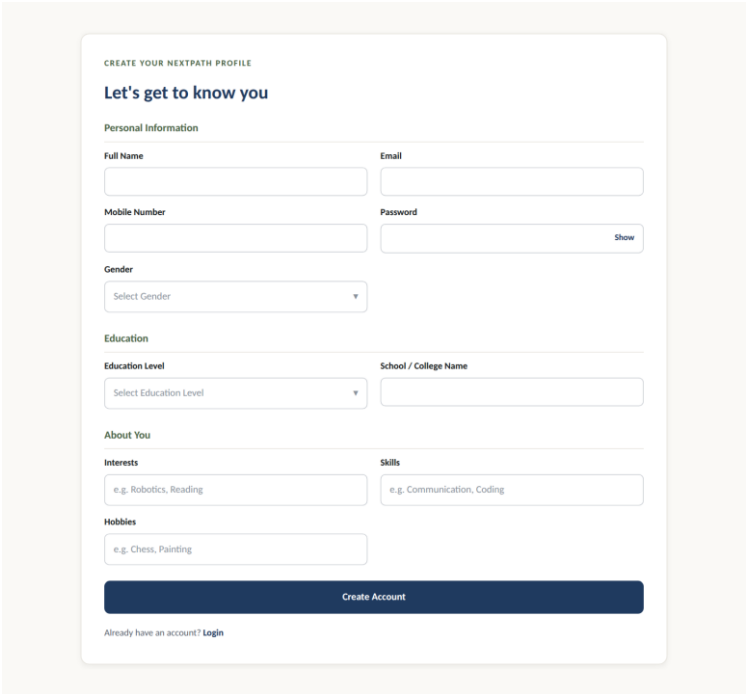


Figure 6.2: Student Registration Page

6.3 Login Page

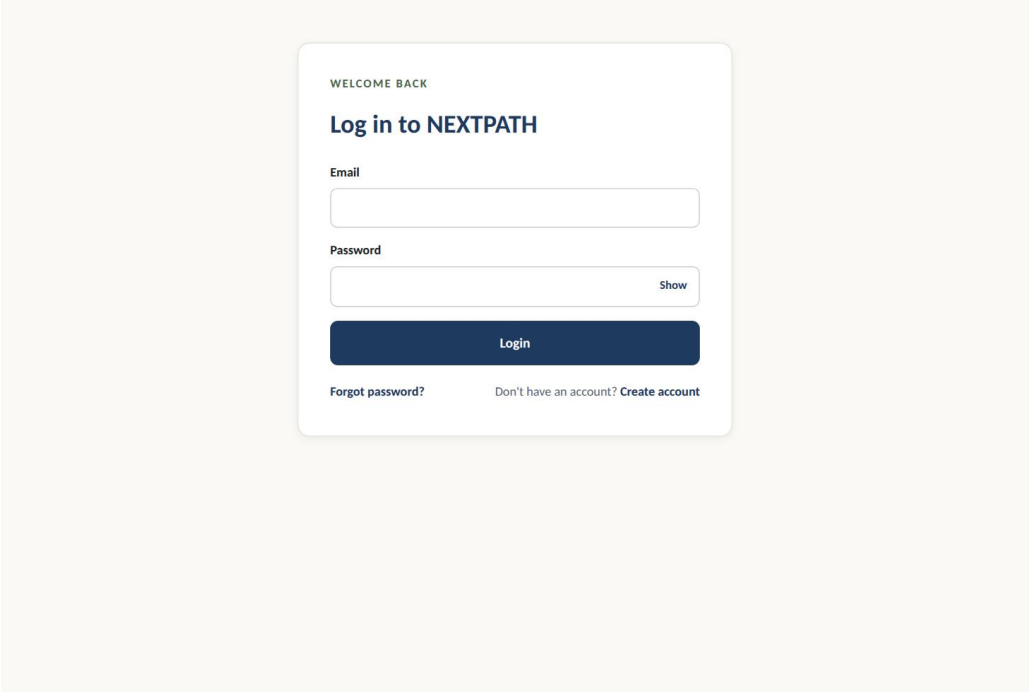


Figure 6.3: Login Page

6.4 Student Dashboard Page

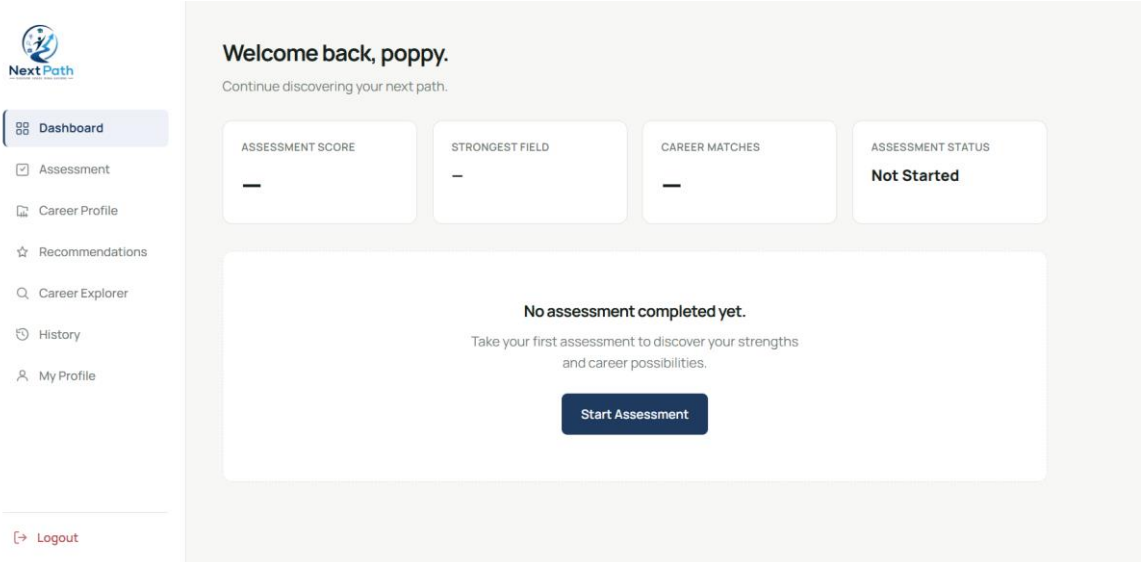


Figure 6.4: Student Dashboard Page

6.5 Student Assessment Page

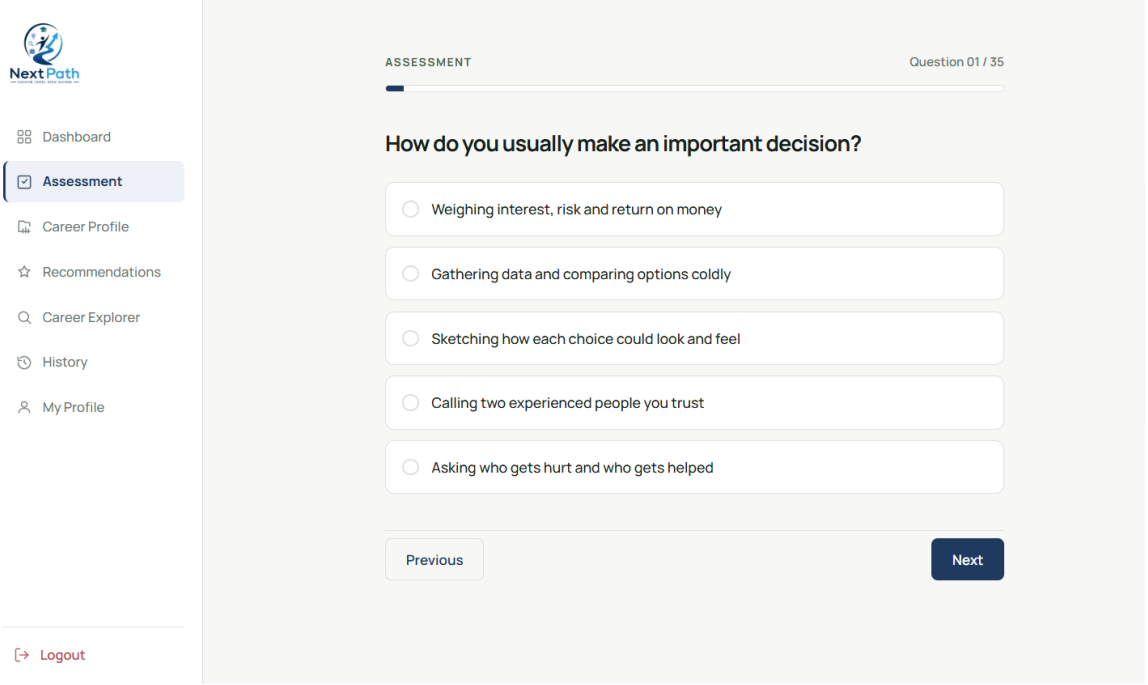


Figure 6.5: Student Assessment Page

6.6 Student Career Explorer Page

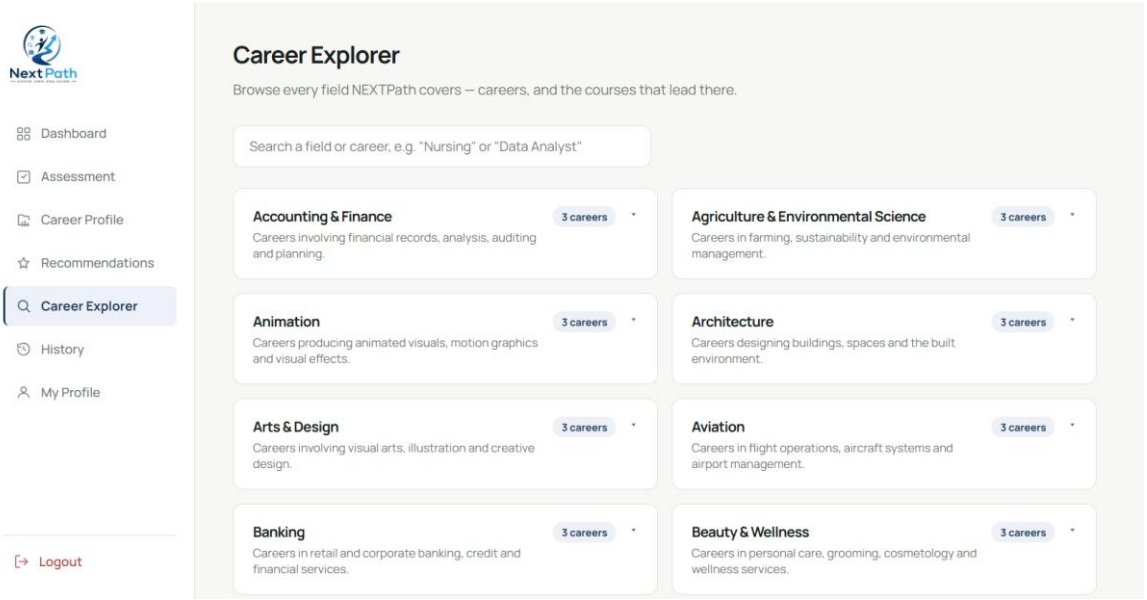


Figure 6.6: Student Career Explorer Page

6.7 Student Assessment Result Page

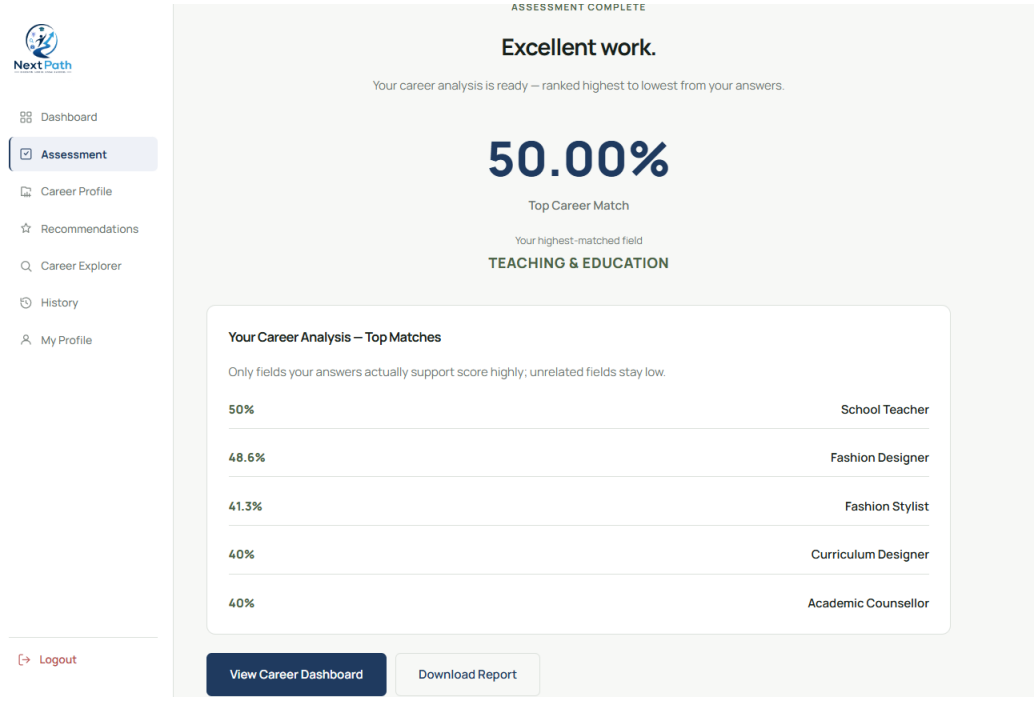


Figure 6.7: Student Assessment Result Page

6.8 Student Career Dashboard Page

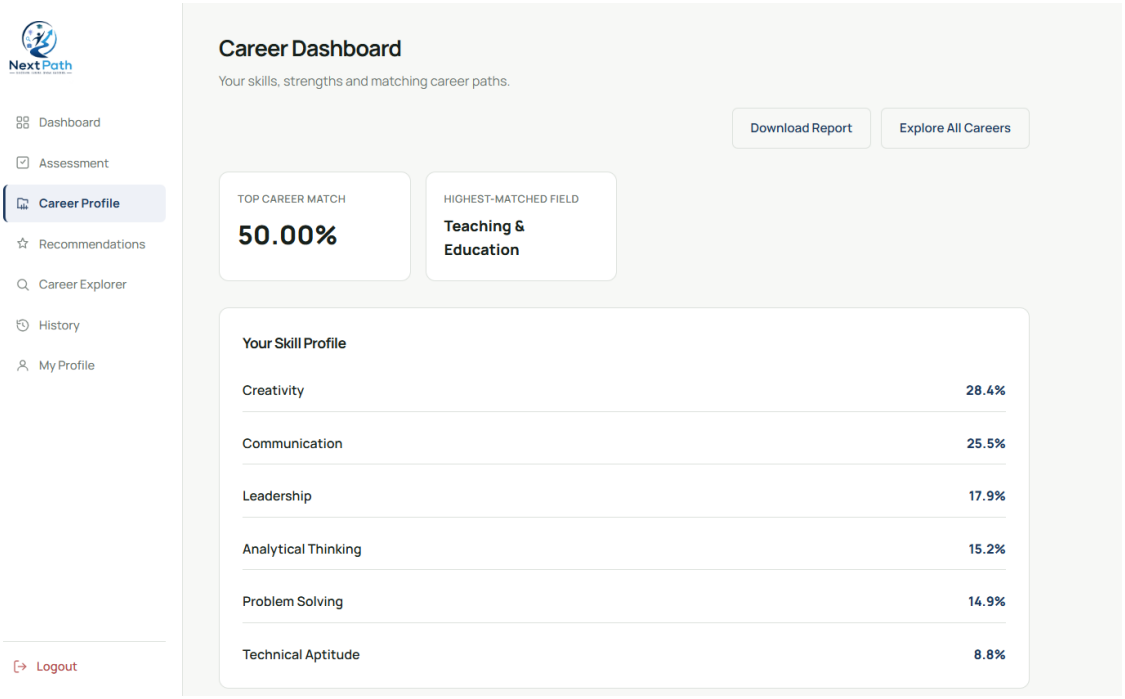


Figure 6.8: Student Career Dashboard Page

6.9 Student Recommended Career Page

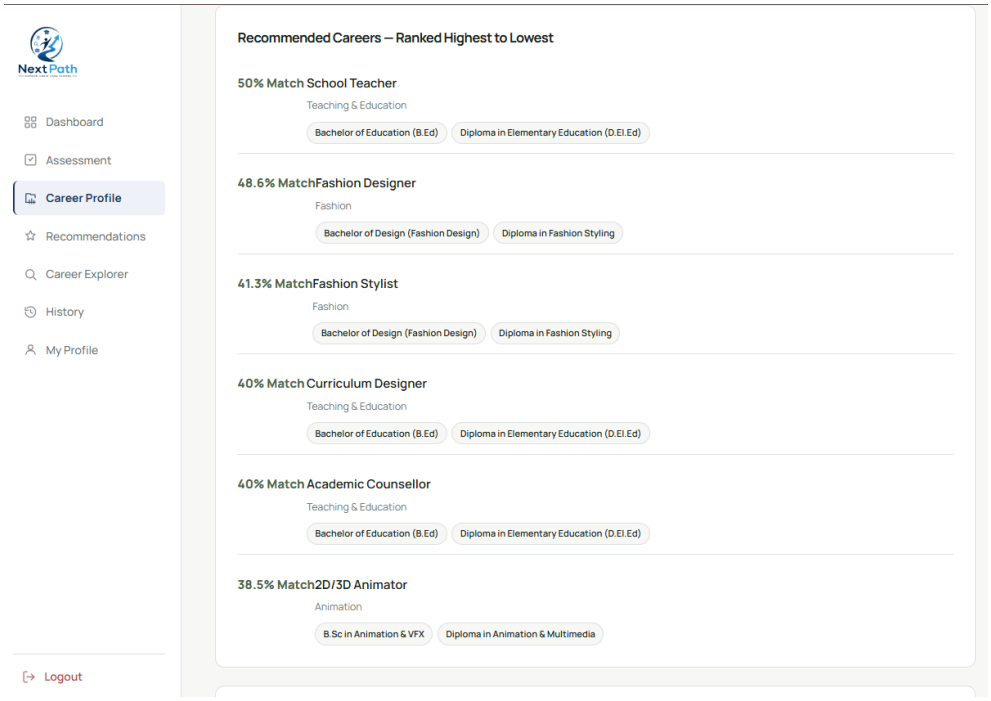


Figure 6.9: Student Recommended Career Page

6.10 Student Assessment History Page

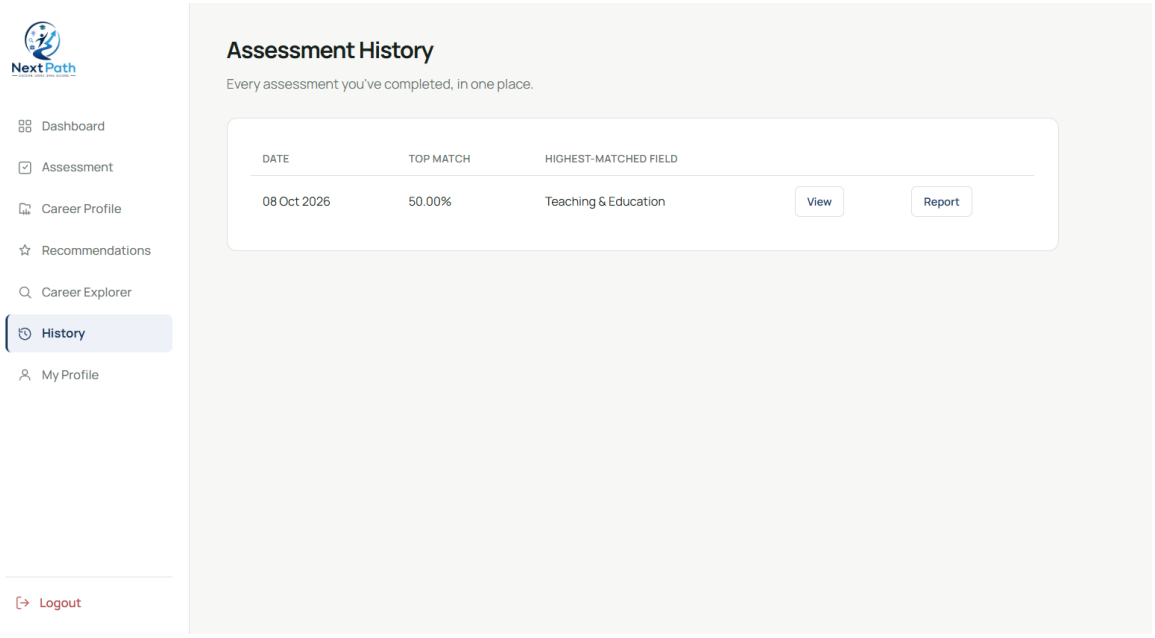


Figure 6.10: Student Assessment History Page

6.11 Student Profile Page

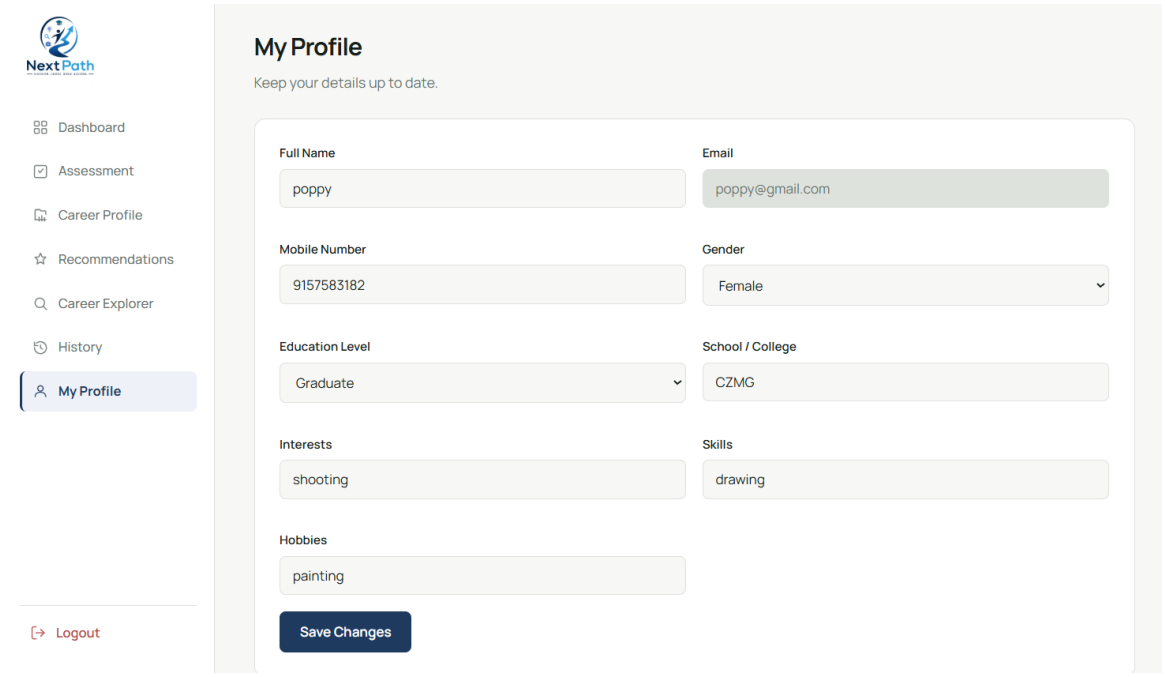


Figure 6.11: Student Profile Page

7. CODE EXPLANATION

7.1 Database Configuration

```php
<?php
/**
 * NEXTPath - Database Connection
 * Uses PDO with prepared statements throughout the application.
 */

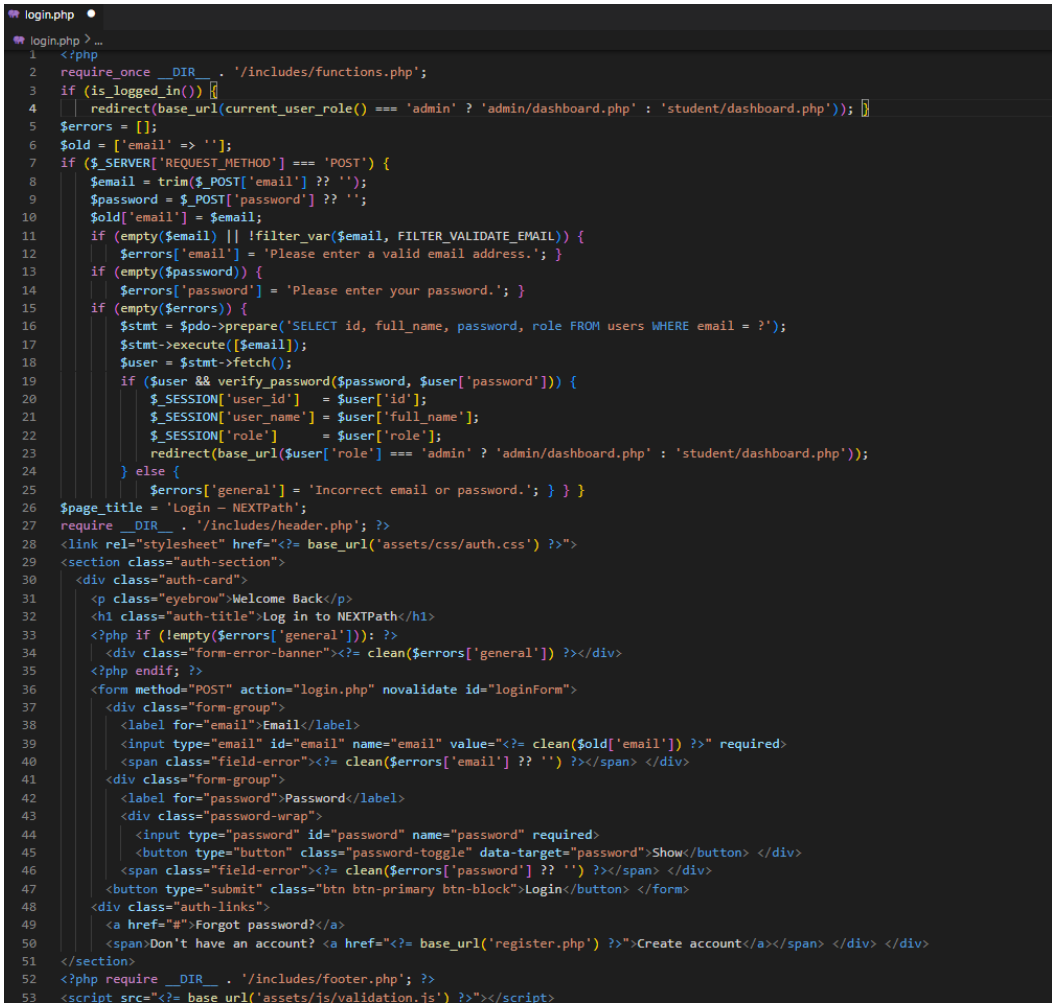
$DB_HOST = 'localhost';
$DB_NAME = 'nextpath';
$DB_USER = 'root';
$DB_PASS = '';

try {
    $pdo = new PDO(
        "mysql:host={$DB_HOST};dbname={$DB_NAME};charset=utf8mb4",
        $DB_USER,
        $DB_PASS,
        [
            PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
            PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
            PDO::ATTR_EMULATE_PREPARES   => false,
        ]
    );
} catch (PDOException $e) {
    error_log('[NEXTPath DB Error] ' . $e->getMessage());
    http_response_code(500);
    die('Something went wrong. Please try again later.');
}
```

Explanation:

1. **Database Connection Setup** – Defines the database credentials (host, database name, username, password) and establishes the connection between PHP and MySQL using new PDO().
2. **Error Handling & Security** – Wraps the connection logic in a try-catch block to hide raw database errors from users and disables emulated prepares (PDO::ATTR_EMULATE_PREPARES => false) for native, secure prepared queries.
3. **Data Fetching Configuration** – Configures PDO::FETCH_ASSOC as the default fetch mode so records are retrieved as clean associative arrays for easier handling in PHP.

7.2 Login Page



```php
<?php
require_once __DIR__ . '/includes/functions.php';
if (is_logged_in()) {
    redirect(base_url(current_user_role() === 'admin' ? 'admin/dashboard.php' : 'student/dashboard.php')); }
$errors = [];
$old = ['email' => ''];
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    $old['email'] = $email;
    if (empty($email) || !filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors['email'] = 'Please enter a valid email address.'; }
    if (empty($password)) {
        $errors['password'] = 'Please enter your password.'; }
    if (empty($errors)) {
        $stmt = $pdo->prepare('SELECT id, full_name, password, role FROM users WHERE email = ?');
        $stmt->execute([$email]);
        $user = $stmt->fetch();
        if ($user && verify_password($password, $user['password'])) {
            $_SESSION['user_id']   = $user['id'];
            $_SESSION['user_name'] = $user['full_name'];
            $_SESSION['role']      = $user['role'];
            redirect(base_url($user['role'] === 'admin' ? 'admin/dashboard.php' : 'student/dashboard.php'));
        } else {
            $errors['general'] = 'Incorrect email or password.'; } } }
$page_title = 'Login – NEXTPath';
require __DIR__ . '/includes/header.php'; ?>
<link rel="stylesheet" href="<?= base_url('assets/css/auth.css') ?>">
<section class="auth-section">
  <div class="auth-card">
    <p class="eyebrow">Welcome Back</p>
    <h1 class="auth-title">Log in to NEXTPath</h1>
    <?php if (!empty($errors['general'])): ?>
      <div class="form-error-banner"><?= clean($errors['general']) ?></div>
    <?php endif; ?>
    <form method="POST" action="login.php" novalidate id="loginForm">
      <div class="form-group">
        <label for="email">Email</label>
        <input type="email" id="email" name="email" value="<?= clean($old['email']) ?>" required>
        <span class="field-error"><?= clean($errors['email'] ?? '') ?></span> </div>
      <div class="form-group">
        <label for="password">Password</label>
        <div class="password-wrap">
          <input type="password" id="password" name="password" required>
          <button type="button" class="password-toggle" data-target="password">Show</button> </div>
        <span class="field-error"><?= clean($errors['password'] ?? '') ?></span> </div>
      <button type="submit" class="btn btn-primary btn-block">Login</button> </form>
    <div class="auth-links">
      <a href="#">Forgot password?</a>
      <span>Don't have an account? <a href="<?= base_url('register.php') ?>">Create account</a></span> </div> </div>
</section>
<?php require __DIR__ . '/includes/footer.php'; ?>
<script src="<?= base_url('assets/js/validation.js') ?>"></script>
```

Explanation:

1. **User Authentication & Validation** – Validates that the submitted email format is valid and the password field is not empty, enabling registered users to log in securely.
2. **Database Verification & Password Hashing** – Searches for the user's email in the users table through a secure prepared SQL statement and verifies the credentials using password_verify().
3. **Session Management & Role Redirection** – Saves the user's ID, name and role into the PHP session upon successful login and redirects administrators to the Admin Dashboard and students to the Student Dashboard.

7.3 Privacy Page

privacy.php

privacy.php > section.section

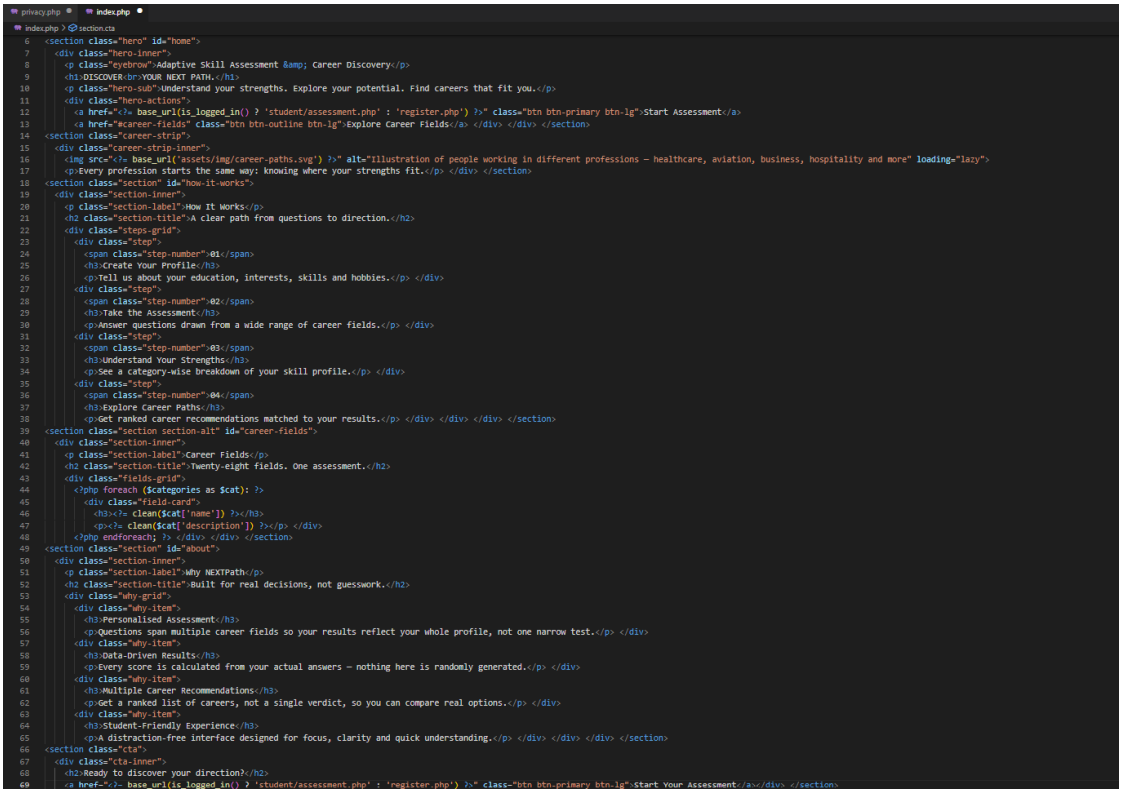
```php
<?php
require_once __DIR__ . '/includes/functions.php';
$page_title = 'Privacy Policy — NEXTPath';
require __DIR__ . '/includes/header.php';
?>
<section class="section">
  <div class="section-inner">
    <p class="section-label">Privacy Policy</p>
    <h2 class="section-title">Your data, handled responsibly.</h2>
    <p style="max-width:640px;color:var(--muted);">
    NEXTPath collects only the information needed to build your skill profile and career recommendations:
    your registration details and assessment answers. We never sell your data, and passwords are stored using one-way hashing.
    Contact us any time to review or remove your data.</p>
  </div>
</section>
<?php require __DIR__ . '/includes/footer.php'; ?>

```

Explanation:

1. **Page Purpose & Structure** – Serves as NextPath's static public privacy policy page, importing functions.php for helper utilities, setting the browser $page_title and maintaining UI consistency through shared header and footer templates.
2. **Data Usage & Protection** – Clarifies that collected information (registration details and assessment answers) is used only to generate skill profiles and career recommendations, with an explicit guarantee that user data is never sold.
3. **Security & User Rights** – Stores passwords using one-way hashing and gives users full ownership to review or request removal of their data at any time.

7.4 Index Page

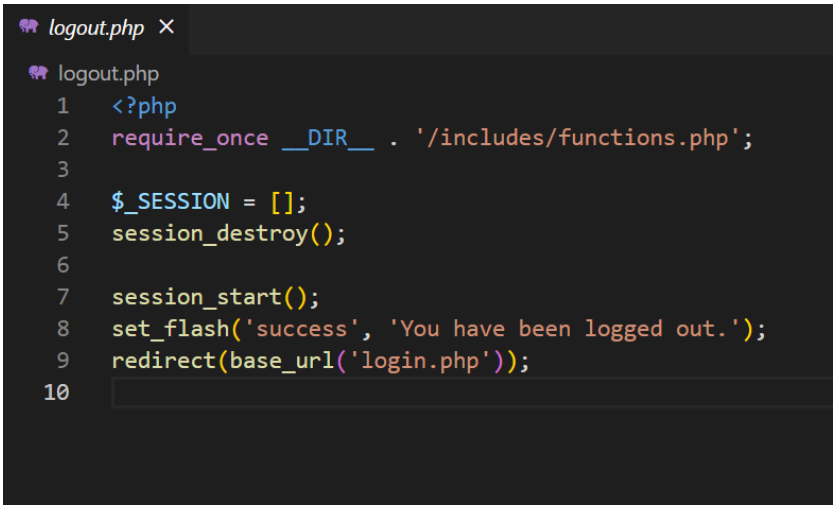


```
privacy.php   index.php
index.php > section.cta
<section class="hero" id="home">
  <div class="hero-inner">
    <p class="eyebrow">Adaptive Skill Assessment &amp; Career Discovery</p>
    <h1>DISCOVER<br>YOUR NEXT PATH.</h1>
    <p class="hero-sub">Understand your strengths. Explore your potential. Find careers that fit you.</p>
    <div class="hero-actions">
      <a href="<?= base_url(is_logged_in() ? 'student/assessment.php' : 'register.php') ?>" class="btn btn-primary btn-lg">Start Assessment</a>
      <a href="#career-fields" class="btn btn-outline btn-lg">Explore Career Fields</a> </div> </div> </section>
<section class="career-strip">
  <div class="career-strip-inner">
    <img src="<?= base_url('assets/img/career-paths.svg') ?>" alt="Illustration of people working in different professions — healthcare, aviation, business, hospitality and more" loading="lazy">
    <p>Every profession starts the same way: knowing where your strengths fit.</p> </div> </section>
<section class="section" id="how-it-works">
  <div class="section-inner">
    <p class="section-label">How It Works</p>
    <h2 class="section-title">A clear path from questions to direction.</h2>
    <div class="steps-grid">
      <div class="step">
        <span class="step-number">01</span>
        <h3>Create Your Profile</h3>
        <p>Tell us about your education, interests, skills and hobbies.</p> </div>
      <div class="step">
        <span class="step-number">02</span>
        <h3>Take the Assessment</h3>
        <p>Answer questions drawn from a wide range of career fields.</p> </div>
      <div class="step">
        <span class="step-number">03</span>
        <h3>Understand Your Strengths</h3>
        <p>See a category-wise breakdown of your skill profile.</p> </div>
      <div class="step">
        <span class="step-number">04</span>
        <h3>Explore Career Paths</h3>
        <p>Get ranked career recommendations matched to your results.</p> </div> </div> </div> </section>
<section class="section section-alt" id="career-fields">
  <div class="section-inner">
    <p class="section-label">Career Fields</p>
    <h2 class="section-title">Twenty-eight fields. One assessment.</h2>
    <div class="fields-grid">
      <?php foreach ($categories as $cat): ?>
        <div class="field-card">
          <h3><?= clean($cat['name']) ?></h3>
          <p><?= clean($cat['description']) ?></p> </div>
      <?php endforeach; ?> </div> </div> </section>
<section class="section" id="about">
  <div class="section-inner">
    <p class="section-label">Why NEXTPath</p>
    <h2 class="section-title">Built for real decisions, not guesswork.</h2>
    <div class="why-grid">
      <div class="why-item">
        <h3>Personalised Assessment</h3>
        <p>Questions span multiple career fields so your results reflect your whole profile, not one narrow test.</p> </div>
      <div class="why-item">
        <h3>Data-Driven Results</h3>
        <p>Every score is calculated from your actual answers — nothing here is randomly generated.</p> </div>
      <div class="why-item">
        <h3>Multiple Career Recommendations</h3>
        <p>Get a ranked list of careers, not a single verdict, so you can compare real options.</p> </div>
      <div class="why-item">
        <h3>Student-Friendly Experience</h3>
        <p>A distraction-free interface designed for focus, clarity and quick understanding.</p> </div> </div> </div> </section>
<section class="cta">
  <div class="cta-inner">
    <h2>Ready to discover your direction?</h2>
    <a href="<?= base_url(is_logged_in() ? 'student/assessment.php' : 'register.php') ?>" class="btn btn-primary btn-lg">Start Your Assessment</a></div> </section>
```

Explanation:

1. **Hero Section & Dynamic CTA** – Introduces the NextPath career assessment platform and uses is_logged_in() to send logged-in users directly to student/assessment.php while routing guests to register.php.
2. **Step-by-Step Workflow & Value Proposition** – Outlines the 4-step assessment journey (profile creation, test, strength breakdown, ranked careers) and highlights core platform benefits such as data-driven scoring and a distraction-free UI.
3. **Dynamic Career Categories & Secondary CTA** – Renders career category cards by looping over $categories with clean() for XSS protection, closing with a bottom call-to-action banner to begin the assessment.

7.5 Logout Page



Explanation:

1. **Loads Helper Functions (require_once)** – Imports essential utility functions from functions.php (such as flash messaging and redirect helpers) using the absolute directory path __DIR__.
2. **Clears and Destroys the Active Session** – Empties the $_SESSION array with $_SESSION = [] and destroys the server-side session data using session_destroy().
3. **Sets a Flash Message and Redirects** – Starts a fresh session with session_start() only to store the confirmation message "You have been logged out." through set_flash(), then sends the user to login.php using redirect().

8. LIMITATIONS AND FUTURE ENHANCEMENT

8.1 Limitations

- The assessment uses a predefined question bank; the quality of the results depends on the questions entered by the administrator.
- Recommendations are rule-based and use stored relevance weights; no machine-learning model is used.
- The career and course data are limited to the data stored in the database (28 categories, 84 careers and 56 courses) and are not updated from external sources.
- Password reset by email is not available.
- Administrator tables do not use pagination, which may be inconvenient with very large data.
- The Career Report depends on the browser's print-to-PDF function.
- The system is a web application only; there is no separate mobile application.

8.2 Future Enhancement

The following are possible future enhancements and are not part of the current system:

- Password reset through email.
- Pagination on large administrator tables.
- Configurable skill-to-category weighting from the Admin Dashboard.
- Export of assessment history to PDF.
- Multi-language support.
- A larger career and course database with more assessment categories.
- Improved or AI/ML-based recommendation logic and personalised learning paths.
- Integration of career trends and external job or internship information.
- A mobile application and notifications.

9. CONCLUSION

NextPath is a secure web-based skill assessment platform designed to help students discover suitable careers.

- **Core Functionality:** Students take a mixed-category multiple-choice test to receive scores, a skill profile, strength analysis and ranked career and course recommendations based on rule-based weighted logic.
- **Admin Module:** Enables dynamic management of questions, categories, careers and courses, along with student records and assessment analytics, without modifying code.
- **Tech Stack & Security:** Built using HTML5, CSS3, JavaScript, PHP (PDO) and MySQL, and secured with password hashing, prepared statements, output escaping and role-based access control.
- **Future Scope:** Expanding the career database, adding advanced analytics, mobile app support and password recovery features.

10. REFERENCES AND BIBLIOGRAPHY

Books

- Luke Welling and Laura Thomson, PHP and MySQL Web Development, Addison-Wesley.
- Roger S. Pressman, Software Engineering: A Practitioner's Approach, McGraw-Hill.
- Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems, Pearson.

Technologies and Libraries

- PHP (with PDO) – server-side scripting language.
- MySQL / MariaDB – relational database.
- HTML5, CSS3 and JavaScript – front-end technologies.
- XAMPP (Apache, MySQL, phpMyAdmin) – local server environment.

Websites

- https://www.php.net/manual – PHP Manual
- https://dev.mysql.com/doc – MySQL Documentation
- https://www.w3schools.com – Web development tutorials
- https://www.apachefriends.org – XAMPP
- https://fonts.google.com – Google Fonts